PHASE 1 — INGEST

Upload Workflow

How your document goes from a raw file to a searchable knowledge base.

1

You upload your file

A PDF or text file is sent through the app, validated, and saved temporarily on the server.

- Multer
- PDF & TXT only

2

The text is read out

All the words are pulled out of the file — like copy-pasting the document into plain text.

- pdf-parse
- UTF-8 reader (TXT)

3

Sliced into small passages

The document is chopped into short, overlapping sections that are easy to search individually.

- Custom chunker
- Size & overlap via config

4

Each passage becomes numbers

An AI model converts every passage into a list of numbers — a fingerprint of its meaning.

- OpenAI API
- text-embedding-3-small

5

Saved in a search library

Numbered passages are stored in a vector database that finds the closest match instantly.

- Pinecone vector DB
- Stored by namespace

Result — your document is indexed and ready to answer questions.

PHASE 2 — RETRIEVE & ANSWER

Ask Workflow

How your question finds the right passages and turns into a sourced answer.

1

Your question becomes numbers

The same AI model converts your question into a fingerprint so it can be compared to every stored passage.

OpenAI API

text-embedding-3-small

2

Most relevant passages found

The database finds the passages whose fingerprints are closest to your question — the ones most likely to hold the answer.

Pinecone similarity search

Returns top-K matches

3

AI reads & writes the answer

Top passages are handed to a language model that reads only those passages and writes a concise, accurate answer — nothing is invented.

OpenAI API · gpt-4o-mini

Temperature 0.2

4

You get an answer with sources

The response comes back with the answer plus exactly which parts of your document it came from — so you can verify it yourself.

Document name & passage

Match score

Result — a grounded answer you can trace back to the source.